

Deep Reinforcement Learning for Traffic Light Control in Jönköping

Ben Bekir Ertugrul
School of Engineering
Jönköping University
Jönköping, Sweden

Ahmet Alper Basturk
School of Engineering
Jönköping University
Jönköping, Sweden

Abstract—TODO

I. INTRODUCTION

Rapid global urbanization and rising populations create a demand for traffic throughput optimization in order to save time, money, and resources, while increasing road safety. Traditional Traffic Signal Control (TSC) systems remain inefficient, relying on static, pre-programmed algorithms that fail to adapt to real-time traffic fluctuations. At the same time, artificial intelligence is experiencing widespread integration across decision-making domains and allows for adaptive, intelligent control systems that offer a direct path toward mitigating urban gridlock.

This paper focuses on the city center of Jönköping, Sweden, as a case study for algorithmic intersection optimization. Jönköping is chosen as its road network experiences severe bottlenecks during peak rush hours, primarily driven by roundabouts feeding directly into closely spaced intersections controlled by traffic lights. Because these existing controllers are based on non-adaptive phase-switching strategies, they are not able to mitigate the compounding queue lengths that propagate across the interconnected junctions. Implementing a Deep Reinforcement Learning (DRL) framework provides an empirical evaluation of how Deep Learning can impact multi-intersection traffic throughput.

A. Structure of this paper

TODO

II. BACKGROUND

A. SUMO

Simulation of Urban MObility (SUMO) is an open-source traffic simulation suite designed to handle large-scale road networks [1]. The platform allows for modeling of streets, vehicles, and traffic signal logics. It incorporates the Traffic Control Interface (TraCI), an API that enables live state retrieval and manipulation of traffic light phases [2]. SUMO also offers fine-grained data logging capabilities for performance metrics, making it highly relevant for intelligent transportation experiments.

B. SUMO-RL

SUMO-RL is a specialized Python library designed to instantiate reinforcement learning environments using the SUMO simulator [3]. It wraps the raw simulation variables retrieved from TraCI API into a standardized interface and provides default function for observation spaces including lane density and queue lengths, and reward functions.

C. Stable Baselines3

Stable Baselines3 offers deep reinforcement learning algorithm implementations built upon the PyTorch framework [4]. It provides optimized architectures for both value-based and policy-gradient methods, including Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO), eliminating implementation bias and serving as a basis for comparative agent analysis.

III. RELATED WORKS

TODO: name 2-3 related works and quickly mention topic, method, and findings

IV. ENVIRONMENT

a good starting tutorial on how to generate traffic scenarios can be found here [5].

A. Network

map of sweden downloaded here:
<https://download.geofabrik.de/europe/sweden.html> geojson polygon object created here: <https://geojson.io/> chosen geojson found here¹.

then, we used osmium and osmfilter to apply the polygon shape to the map of sweden. afterwards, we filter only for highways, except for those of the following types:

- footway
- path
- pedestrian
- steps
- cycleway
- bridleway

finally, we use the SUMO *netconvert* tool to convert the map into a SUMO network.

¹<https://github.com/benbekir/traffic-rl/blob/main/simulations/networks/jkpg/artifacts/jkpg.geojson>



Fig. 1. The filter (in grey) that defines the boundaries of the road network in Jönköping.

as for limitations, we extract lanes including speed limit, intersections, and traffic lights. however, since we only model car traffic and not pedestrians, all pedestrian traffic lights will behave completely deterministic. this does not represent the real world where pedestrians can press a button on the traffic light to trigger a faster phase switching.



Fig. 2. The network as shown in the SUMO nededit UI, highlighting the traffic light placements.

1) *Demand Modeling*: since there is no publicly available data regarding traffic assignment zones and demand in jönköping, we use the SUMO tool randomTrips² to randomly create trips. we create 3600 individual trips in total where a new trip is spawned into the simulation on every timestep.

2) *Rush Hours and Noise*: im not sure about this yet

B. Observations

- lane density: density is calculated on the entire lane, by considering how many cars are present in the lane and how many cars could potentially fit in the lane. https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L297

²<https://sumo.dlr.de/docs/Tools/Trip.html#randomtrips>

- queue information: queue is calculated by counting the number of cars that are stopped at the traffic light, waiting for the green signal. https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L309

C. Actions

D. Reward

- predefined reward functions: https://github.com/LucasAlegre/sumo-rl/blob/main/sumo_rl/environment/traffic_signal.py#L347

V. EXPERIMENT SETUP

A. Proximal Policy Optimization

as a policy-gradient method, proximal policy optimization (PPO) learns only from the recent experiences that the agent collected.

B. Deep Q Networks

A Deep Q-Network (DQN) is a reinforcement learning algorithm that combines traditional Q-learning with deep neural networks.

TABLE I
OVERVIEW OF THE TRAFFIC SIGNAL CONTROL SYSTEM.

Parameter	Description
Method	Deep Q-Network (DQN) and Proximal Policy Optimization (PPO)
Technique	Curriculum Learning with three traffic density stages
Scenario	(0.75, 0.875, and 1.0 scale factors) Real-world road network of Jönköping, Sweden
Platform	Simulation of Urban Mobility (SUMO)

TABLE II
OVERVIEW OF THE TRAFFIC SIGNAL CONTROL SYSTEM.

Parameter	Description
Method	Deep Q-Learning (DQN) and Proximal Policy Optimization (PPO)
Technique	Curriculum Learning with parameters and three traffic density stages (0.75, 0.875, and 1.0 scale factors)
Scenario	Jönköping city center road network (multi-intersection)
Platform	SUMO (Simulation of Urban Mobility)
Agent Type	Multi-Agent with shared policy (parameter sharing)
Reward Function	Queue-based penalty

Ben, choose one of the graphs.

VI. RESULTS

TABLE III
PERFORMANCE COMPARISON ACROSS 10 RUNS (AVERAGED).

Metric	Baseline	DQN	PPO
Completed trips	5,837	5,192	5,443
Avg. waiting time	97.4s	12.5s	35.1s
Avg. time loss	199.4s	45.2s	89.3s
Avg. stops per vehicle	6.48	1.11	2.97
Avg. speed	6.73 m/s	8.81 m/s	8.10 m/s
Total waiting time	568,322s	64,689s	201,287s
Simulation end time	8,398s	8,398s	8,157s

TABLE IV
IMPROVEMENT OVER BASELINE (10-RUN AVERAGE).

Metric	DQN	PPO
Waiting time	-87.2%	-64.0%
Time loss	-77.3%	-55.2%
Stops per vehicle	-82.8%	-54.2%
Average speed	+30.9%	+20.5%
Completed trips	-11.0%	-6.8%

VII. CONCLUSION

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